

Online retail customer behavior

What the transaction data can tell leadership about revenue, loyalty, cancellations, and seasonal demand.



UK online gift-ware
retailer

Dec 2010 - Dec
2011

Commercial readout

Four decisions sit behind the analysis

Frame the deck as a decision readout,
not a tour of the tooling.

Which markets and products deserve commercial focus?

Which customers create repeatable value?

Where are cancellations eroding demand and revenue?

How should marketing and inventory follow seasonal patterns?

The dataset is broad enough for both market and customer questions

Eight columns connect invoice, product, date, price, customer, and country.

541,910

Raw line items across 38 countries

531,392

Rows after cleaning

4,372

Unique identified customers ·
approximately 4,070 products

8 Columns

Invoice, product, date, price,
customer, and country

Cleaning preserved business signal instead of hiding it

Data cohort	Governance decision & impact
Missing product descriptions	Recovered for 1,342 of 1,454 incomplete rows using StockCode lookups
Customer ID	135,080 rows lacked a Customer ID, or 24.9% of raw data. Those unidentified transactions were retained for revenue and product analysis, then excluded from customer-level RFM and loyalty work
Duplicates, non-product rows, and invalid prices	Removed
Cancellations and returns	Stayed in the data

Revenue needs to be read on three levels

Market level Geography & order value	Product level Catalog performance & concentration	Time level Seasonal monthly cadence
UK domestic · 82%+ total sales	Catalog breadth · ~4,070 items	H1 baseline · Steady run-rate
International export · 37 nations	Top 5 products · Disproportionate gross	Q3 build · Pre-holiday stock
Non-UK AOV · Higher export basket	Country variance · Regional demand	Q4 surge · Oct–Nov peak revenue
Priority · Target core Europe	Priority · Protect hero SKUs	Priority · Stage inventory early

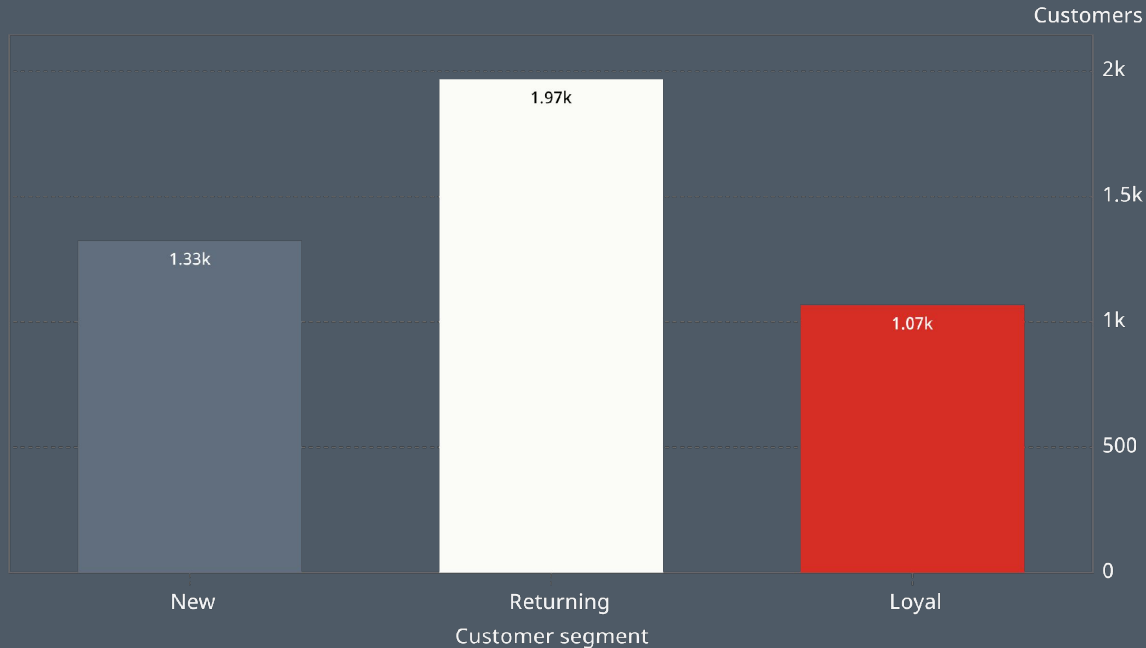
Customer value is more useful than customer count

Customer analysis

- Segment identified customers into New, Returning, and Loyal groups based on distinct orders.
- Test whether customers with more than five orders spend more on average than non-repeat buyers.
- Isolate high-spending returning customers who have experienced at least one returned order.

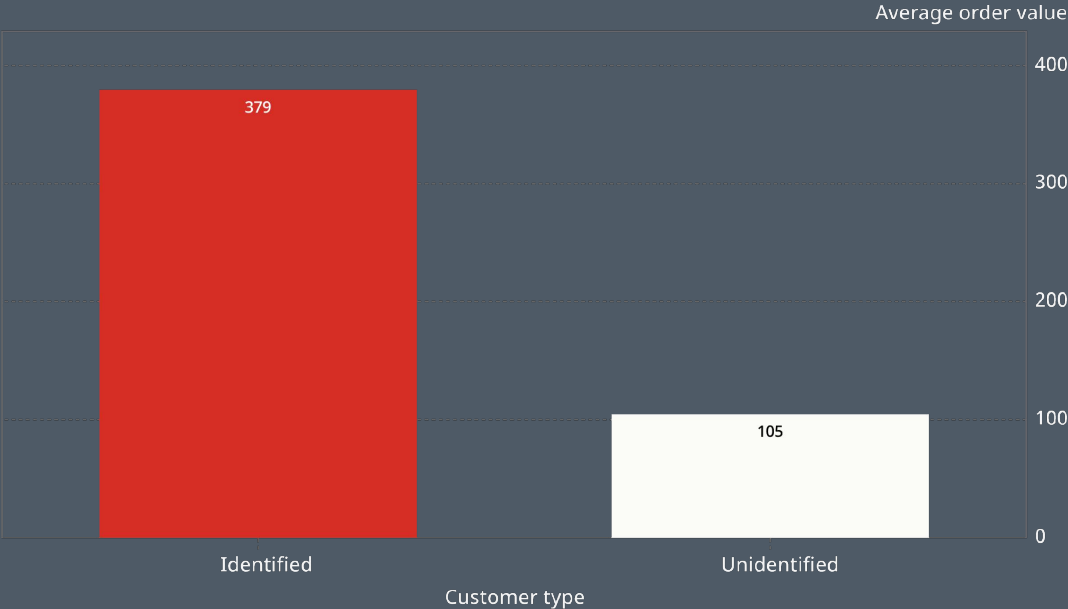
Retention directive

Retention should be managed by value and behavior, not only by volume.



Guest and wholesale activity should be treated as its own commercial lane

Unidentified transactions135,080 rows · 24.9% of total	Identified customer accounts396,312 clean rows · 4,372 accounts
Transaction structure · No registered Customer ID	Transaction structure · Verified CustomerID profile
Purchasing profile · Guest checkout & wholesale bulk	Purchasing profile · Trackable individual & retail buyers
Order economics · Distinct basket sizes & bulk quantities	Order economics · Predictable lifecycle and cadence
Reporting role · Essential for gross revenue and inventory	Reporting role · Primary base for RFM and retention scoring



Cancellations are a performance signal, not a cleanup detail

Concentrated catalog friction

Specific product lines exhibit extreme cancellation rates, pointing to quality discrepancies, packaging fragility, or inaccurate catalog listings.

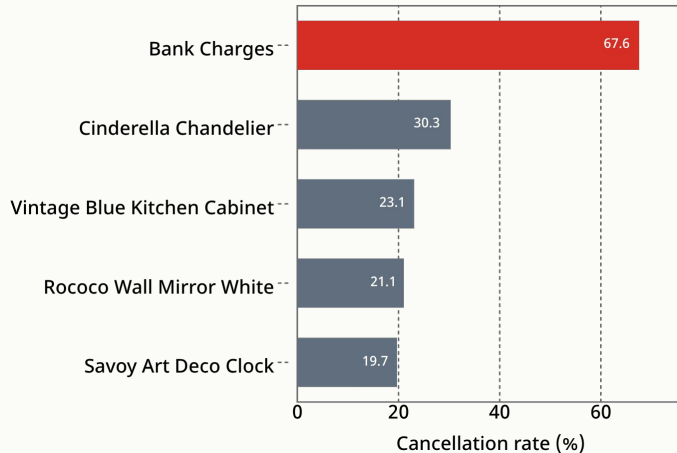
Fulfillment & inventory strain

Cancelled orders tie up warehouse stock during peak trading cycles, inflating ghost inventory and generating reverse-logistics handling overhead.

Executive KPI visibility

Tracking cancellation rate as a primary executive metric transforms operational waste into an active governance review.

Product



Commercial impact

Direct Margin Erosion

Excluding cancellations artificially flatters revenue figures. Feeding cancellation rates back into supplier quality reviews and merchandising eliminates structural fulfillment drag.

Seasonality should change the operating calendar

Revenue by month

December 2010–December 2011

Calculated across the full period.

The dashboard pairs

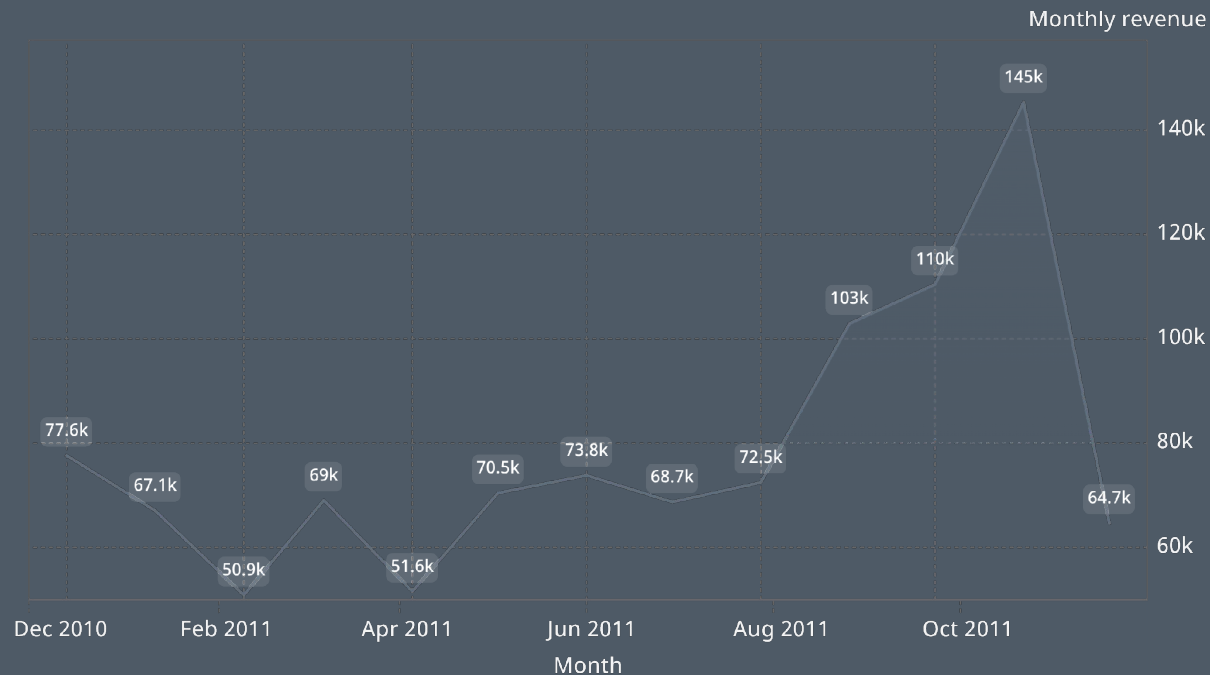
Monthly revenue

With top products by quantity sold.

One leadership view

Timing · assortment · stock

planning
Information can connect timing, assortment, and stock planning in one leadership view.



The monthly series is the bridge from descriptive analysis to campaign and inventory timing.

Power BI turns ten SQL questions into one operating view

KPI cards

- Total revenue
- Number of customers
- Average order value
- Cancellation rate

Customer mix

- Identified versus unidentified orders
- New / Returning / Loyal segments

Commercial views

- Revenue by country
- Revenue by month
- Top products by quantity